



Data's environmental impact: The hidden costs of datafication, storage and analysis

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Learning objectives

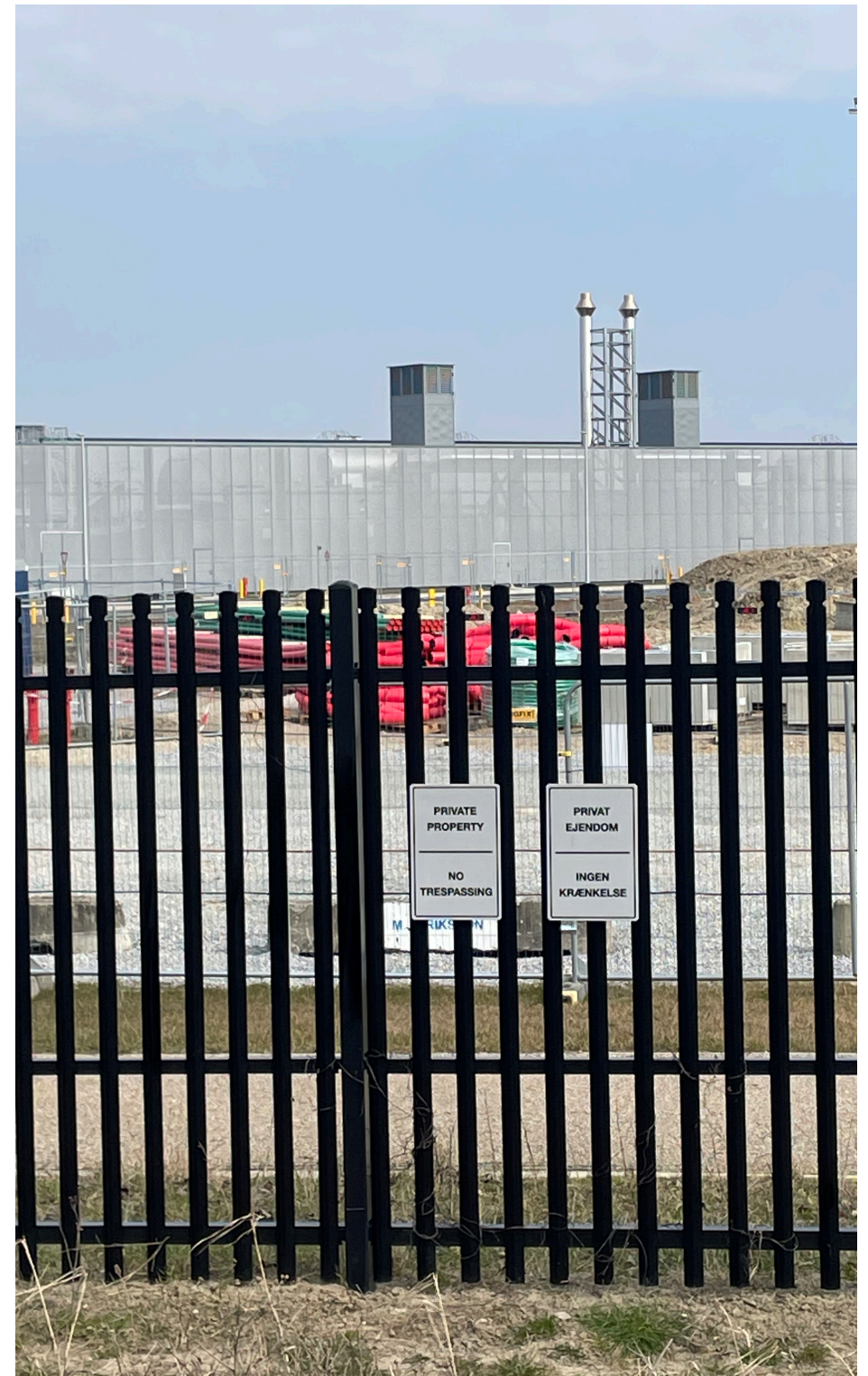
- Understand data as a material system
- Identify environmental impacts
- Apply *supply chain capitalism* and *algorithmic harm*
- Reflect on responsibility issues

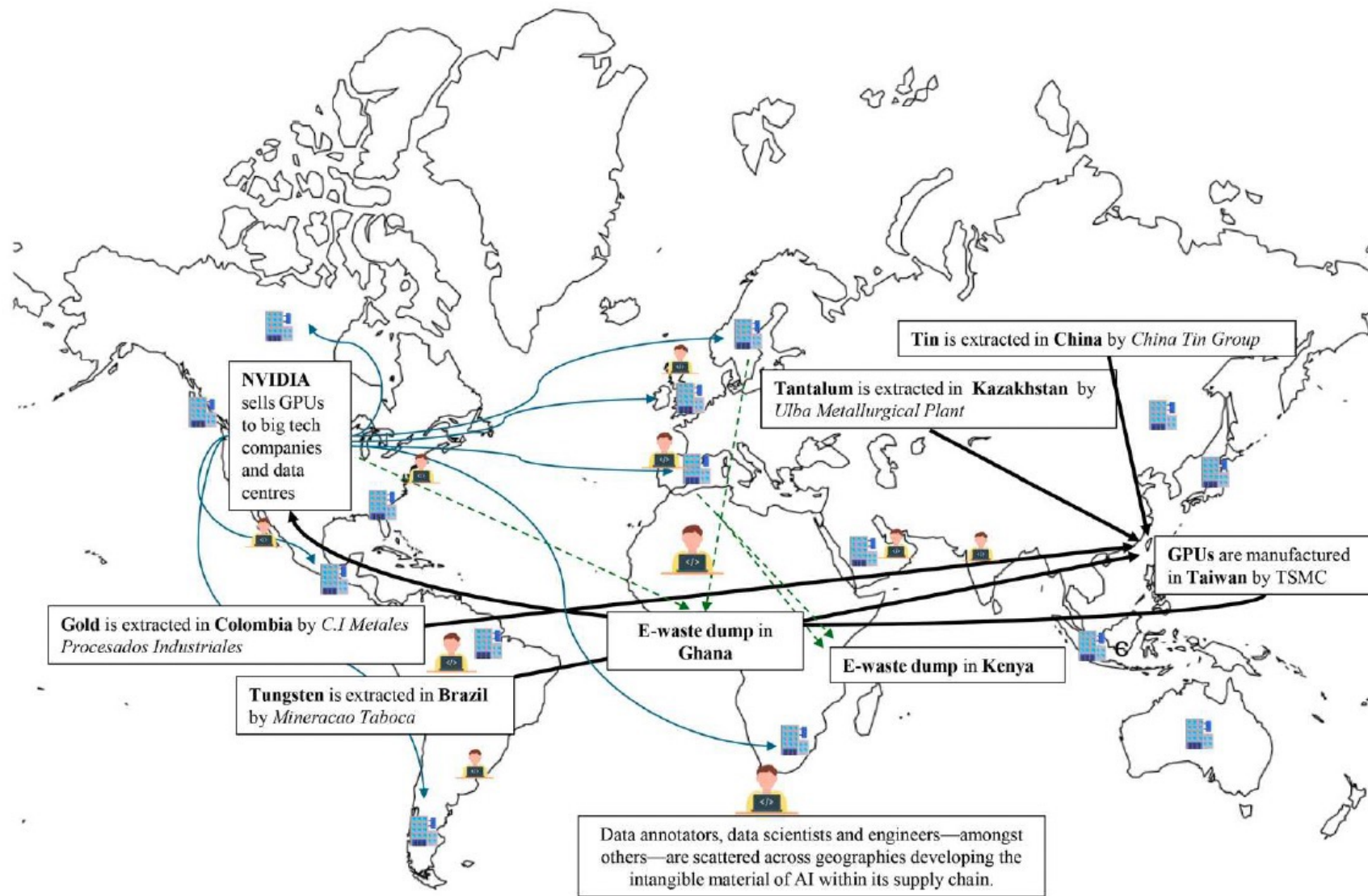
Opening question

- Where is your data right now?
- Why would you bother?



Data is always produced,
processed, and placed
somewhere





Source: Valdivia, Ana. 2024. “The Supply Chain Capitalism of AI: A Call to (Re)Think Algorithmic Harms and Resistance through Environmental Lens.” *Information Communication and Society*.

Quiz 1 – Supply chain capitalism

- The meaning of the concept?
 1. Capitalism works the same way everywhere
 2. Firms control all production directly
 3. Global capitalism is made up of very different labour, laws, and conditions
 4. Technology removes inequality from markets

Quiz 1 Debrief

- Low cost and flexibility is built into the system, but there are other costs
- Low paid workers carry risk while firms capture value
- Heterogeneity not equal to fairness

Key tension in data's supply chain

- Collection
- Organization
- Processing
- Storage
- Delivery

➤ Global vs local

➤ Competing interests and values

Anthropological study

- Apple in Viborg (2017-2020)
- Meta in Odense (2019-2023)
- Microsoft in Eastern and Western Denmark (2024-)

Salling, Caroline Anna, and Brit Ross Winthereik. 2025. "Inventorying Data: Circumvented Investment Conditions by Big Tech's Supply Chain Capitalism." *Journal of Cultural Economy*.

Maguire, James, and Brit Ross Winthereik. 2021. "Digitalizing the State: Data Centres and the Power of Exchange." *Ethnos* 86 (3): 530–51.

Hyperscale data centers - size

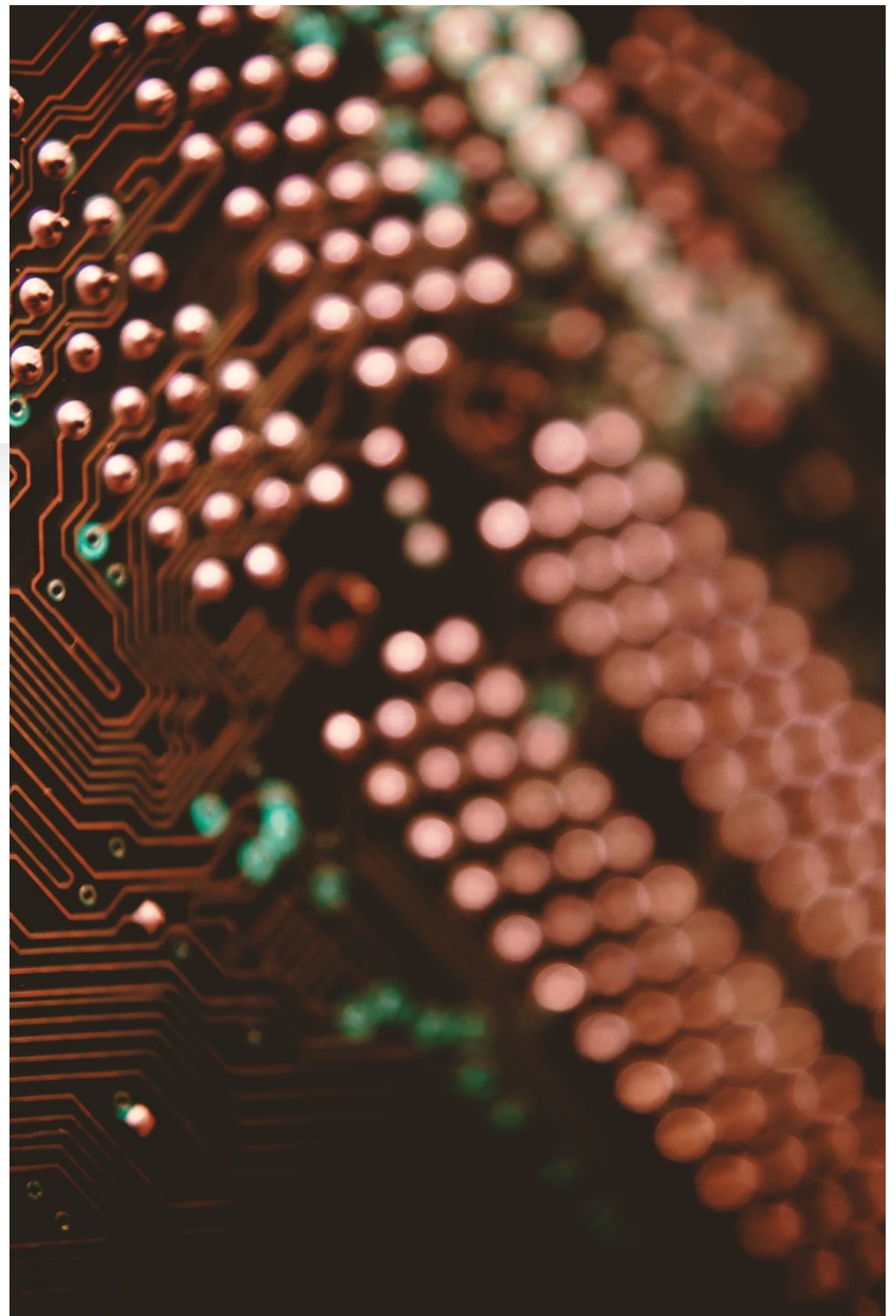
- 5000+ servers
- Minimum of 10.000 square feet/930 square meters/1 ha
- Minimum 40 MW of capacity

Hyperscale data centers - capacity

- Elastic computing: The ability to quickly expand or decrease computer processing, memory, and storage resources to meet changing demands
- Hyperscalers mirror data by maintaining exact copies across multiple locations

Key insight

- Data centers form the infrastructural backbone of the digital society, but their operation puts pressure on local environments



Environmental impact

- Water usage – emptying of aquifers, rising prices for residents
- Energy consumption – pressure on electricity infrastructure, putting climate action to a halt
- Air pollution – reliance on back up diesel generators
- Noise pollution – humming servers
- Light pollution – security

QUILI.AI

THE STORY OF A
COMMUNITY ON A MISSION
TO SAVE ITS WATER.

Quiz 2 – Algorithmic harm

What is algorithmic harm?

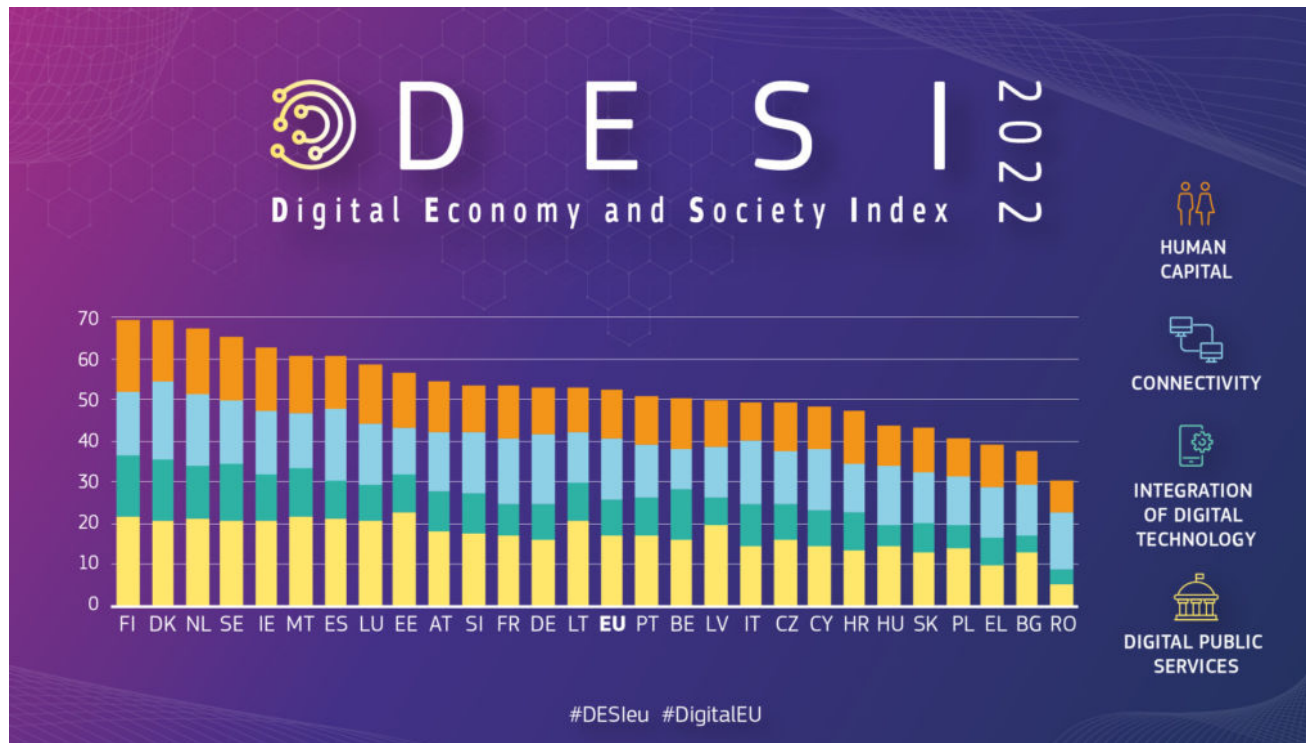
1. When algorithms are trained on biased data
2. When the automation of decisions cause harm to people, because the data is biased
3. When the use of algorithms cause harm to individuals, communities and environments
4. When algorithms make mistakes

Quiz 2 Debrief

- Algorithmic impact is distributed
- It is about the data and bias in data, but not only
- Algorithmic harm must be considered across data's supply chain
- Quili.ai as an example of distributed impact that has negative consequences

Case: Denmark

- Highly digitalized
- Data center expansion due to cool climate and security concerns

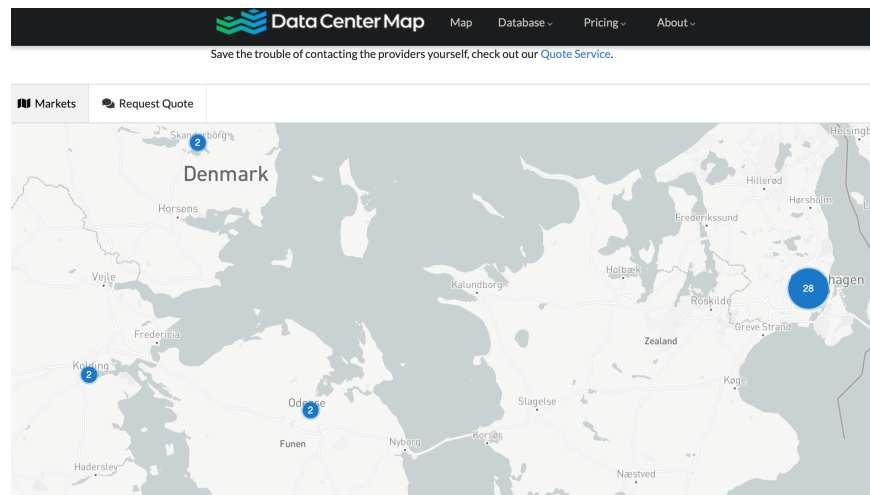


Data centers in Denmark

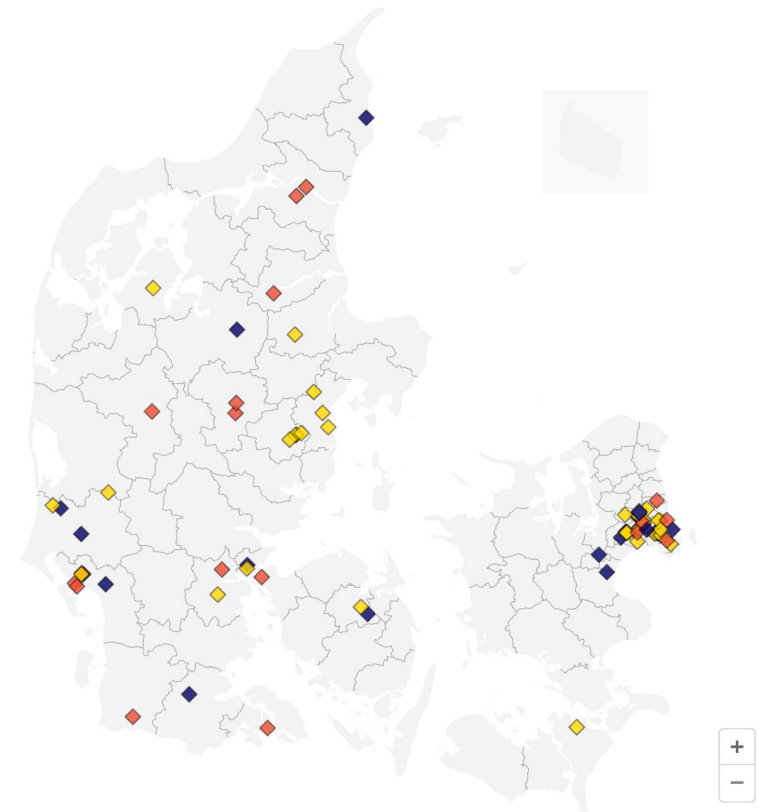
Datacentrene samler sig i hovedstadsområdet og Vestjylland

Zoom ind på kortet for at se, hvor datacentrene ligger i kommunerne

Amerikansk Dansk Europæisk Andet



70 hyperscalers and 12 in process



Dækker colocation-, hyperskalerede- og operatørdacentre. Kortet inkluderer også datacentre, der er planlagte eller under konstruktion.

Kort: Caroline Brundle Bugge • Kilde: [Data Center Map](#), Datacenter Industrien

Promises by the industry

- Jobs
- New investments due to the “AI revolution”
- Data security

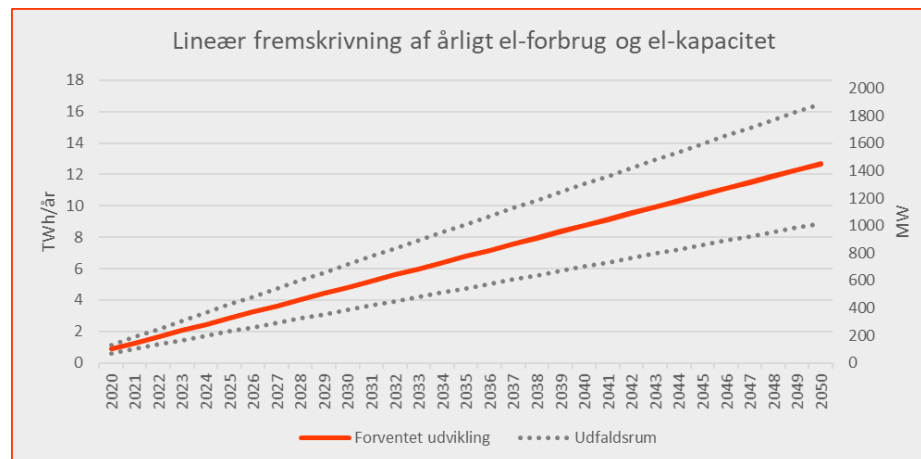
Job creation

Apple's data center, 300 jobs during construction (government estimate), 80 operational jobs expected

Reality after opening in 2020: Only a few small companies moved to the area, Apple initially planned 50 operational staff, with a long-term plan of up to 150

By 2025: 8 employees at the data center

Concerns regarding pressure on the energy infrastructure



»Det kalder på politisk stillingtagen«: Ministeren frygter, at amerikanske techgiganter kommer foran danske virksomheder

I et interview med Finans luftter energiminister Lars Aagaard (M) nu en ny bekymring. Risikoen for, at USA's techkæmper får særlig adgang til elnettet, skal undersøges - og danske virksomheder skal prioriteres.



Arkivfoto: Jens Dresling/Thomas Lekfeldt. Collage: Anders Thykier

ERHVERV 23. JAN. 2026 KL. 11:35

FØJ TIL LÆSELISTE

In DK min. 55% of the energy supply must come from renewable sources. If the demand increases due to data centres, and renewable energy supply cannot be secured, it will threaten society's green transition.

AI and productivity gains

- Analysis from the Danish National Bank
- “Examples of large productivity gains from using AI for certain types of tasks, such as programming and translation. Several international studies predict that AI will also contribute positively to productivity growth in the economy as a whole in the coming years. However, **the extent of the effect is uncertain and varies across countries**. It is also uncertain when the impact of AI will be realised in productivity figures.

Source: <https://www.nationalbanken.dk/en/news-and-knowledge/publications-and-speeches/analysis/2026/artificial-intelligence-can-boost-productivity-in-the-danish-economy>

Water

- Hyperscalers use water for cooling of servers
- They use drinking water
- In DK the drinking water supply is under pressure due to effects of industrial agriculture (PFAS and other substance)

Excess heat

- Hyperscalers can and have been connected to district heating to reduce carbon emissions.
- The heat they produce must be treated further
 - public investment
- The supply is not guaranteed

The citizen perspective

- Citizens have so far not been provided with much information
- Placement is a “technical issue” that removes sustainability and other issues away from public scrutiny
- 2026: Citizen meetings and hearings across Denmark

Hearing responses - a range of emotions

Anxiety

"meget røvrendte" (feeling screwed over)

Frustration

"tromlet over" (steamrolled), Microsoft "ligeglade med os" (doesn't care about us)

Powerlessness

"den store fejl" (the big mistake)

Indignation

"SKAM JER!!!!" (multiple exclamation marks)

Sadness

"knust indeni" (crushed inside)

Loss

"livsværk ødelægges" (life's work destroyed), 40 years of farming, "mit hjem" (my home), home as "resultat af mange års arbejde" (result of many years' work)

Fear

"søvnløse nætter" (sleepless nights), "general belastning" (general burden)

Sociotechnical lens

- Data center issues are composed by social, technical and political elements
- **Engineering work is embedded in global power relations**
 - Architecture
 - Infrastructure
 - Efficiency
 - Data use???

Quiz 3 - Responsibility

- Why does supply chain capitalism make responsibility harder to assign?
 1. Products are too complex
 2. Firms outsource production to legally distant suppliers
 3. Workers choose unsafe jobs freely
 4. Governments control everything

Quiz 3 Debrief

- Best ways to reduce harm induced by data centers?

Reminder

- Technology is socially embedded
- Data is infrastructurally distributed and physically located
- Datafication, storage and data analysis negatively impacts environments

Reflection

- How would you design differently?

